

Global Agrifood Value Chains, Structural Transformation and Income Inequality

Abstract

Global agrifood value chains (GAVCs) are rapidly expanding in many countries around the world and are crucial in shaping various stages of agricultural production and food system transformation. Despite this importance, their socioeconomic implications are not well established. We examine the relationship between GAVCs and income inequality. We construct a rich panel of 167 countries over a 31-wave period 1990–2020 from the EORA Multi Region Input-Output (MRIOs) tables sourced from the UNCTAD-Eora GVC database and other databases. We show a negative association between GAVCs and income inequality; in response to greater GAVC participation, inequality declines. This result is largely driven by both the downstream food and beverage and upstream agricultural sub-sectors. We underscore substantial geographical and income group heterogeneities. Delving into potential explanations for these findings, we show that participation in downstream food and beverage GVCs is associated with a lower share of informal output and self-employment. This is to say that participation in the food and beverage sub-sector may reduce informality and to some extent increase formal jobs in the industrial and service sectors which indicates some aspects of job quality. We also show that structural transformation could explain these findings. While participation in the agriculture GVC increases employment in agriculture, it reduces employment in the industrial and service sectors. Participation in the food and beverage sub-sector, on the other hand, reduces employment in agriculture and pushes up employment in the service and industrial sectors.

Keywords: Agrifood systems; global value chains; income inequality; labor market; structural transformation; job creation

JEL codes : C23, D31, D33, O13, Q17

1. Introduction

Global agrifood value chains (GAVCs) have been rapidly expanding in many countries around the world and are crucial in shaping various stages of agricultural production and food systems transformation. They act as vehicles whereby intermediate inputs are traded to produce final goods over different geographical countries (regions). Agrifood value chains link rural farmers with urban (food) consumers but have been understudied especially from a global perspective (Barrett et al. 2022). Many countries are increasingly been connected through global production networks, with products crossing borders multiple times through different kinds of vertical and horizontal integration (Sexton 2013; Balié et al. 2019). An example of this is Nestlé which is headquartered in Switzerland and produces different products sourcing its various products from different parts of the world. For instance, hazel nuts come from Türkiye, coconut from Sri Lanka, the Philippines and Indonesia, sugar cane from Mexico and Thailand, vegetables from France, meat and poultry from the United States and Germany, spices from India and Vietnam, oil palm from Indonesia and cocoa and coffee from West Africa.

This new configuration of international trade and global production characterized by the move of multinationals to the global arena through production fragmentation, offshoring and outsourcing has positively impacted several countries' income (Goldberg and Pavcnik 2007; Harrison et al., 2011; Helpman et al., 2017). GVCs participation allow countries to benefit from the comparative advantages of other countries. They have been shown to induce technological progress (Wang et al. 2021a; Nana, 2022), increase firm and country level productivity (Amiti and Konings 2007; Pahl and Timmer 2020), generate rising markups (Loecker et al. 2016), and lead to growth and structural transformation (Goldberg et al. 2010; Sampson 2016; Sunghum 2021). Consequently, they represent an emerging opportunity in recent years in not only connecting emerging market and developing economies with global production hubs, but also raising their competitiveness in world trade (IMF, 2016; WTO, 2014). However, it also constitutes a potential threat for inequality and may increase the transmission of

economic shocks (Cattaneo et al., 2010; Shepherd, 2013; Ha et al., 2019; Lopez Villavicencio et al., 2021).

There exist evidence that global value chains have led to increasing income streams (van den Broeck et al. 2017) with implications for economic welfare and development (Pahl and Timmer 2020; Dünhaupt and Herr 2021). Some extant literature have highlighted the link between GVCs participation and income inequality, as GVCs could lead to more pronounced distribution of wage incomes within countries (Antras 2020, Wang et al. 2021b). This relationship is akin to and evokes insights from some theorems in traditional and modern international trade that trade integration could lead to wage inequality (Goldberg and Pavcnik 2007, Helpmann et al. 2017). Following this, GVCs may increase income inequality in less developed countries, although the impact may depend on the type of GVCs participation and several additional factors including the magnitude of the participation, sectors driving such an increase and the position of countries in these GVCs.

Thus, the relationship between GVCs participation and inequality could also be negative since it is expected that income gains from GVCs are likely to trickle down to everyone and reduce inequality (Goldberg and Pavcnik 2007; van den Broeck et al. 2017; Carpa and Martínez-Zarzoso 2022). GVCs have also been associated with increasing employment shares and job creation effects (Banga 2016; Farole 2016; Sunghum and Kim 2021) resulting in significant labour market implications (Lee and Yi 2018). In addition, several developing countries have long been exporters of raw materials, with an important share of the workforce employed in the agricultural sector. An improvement in the level of GAVCs participation, might improve wages and living condition of an important share of the population employed in the agricultural sector and lead to a reduction of income inequality. Despite some attempts, empirical insights on the relationship between GVCs participation and inequality is not well understood especially in the light of structural transformation and wage employment.

In this study, we examine the relationship between GAVCs and income inequality. We look at both the mainstream agricultural and food sectors. Here, we examine both the upstream and the downstream segments of the value chains—also dubbed forward and backward linkages, respectively. To uncover the drivers of this relationship, we examine the role of informality as a naïve proxy of job quality (Hovhannisyan et al. 2022) and some aspects of structural transformation (Lim 2021). We rely on a constructed panel of 167 countries over 31 years (1990–2020). The panel data are constructed from the EORA Multi Region Input-Output (MRIOs) tables sourced from the UNCTAD-Eora Global Value Chain (GVC) database. This is merged with inequality data (Gini) from the World Income Inequality Database (WIID)¹ hosted by UNU-WIDER. We apply different empirical strategies and methods including country fixed effect regressions, instrumental variable (IV) estimators exploring both the standard IV approach and the Bartik shift share IV design, and the generalized method of moments (GMM).

We find a negative relationship between GAVCs and income inequality; that is, income inequality decreases in response to greater participation in GAVCs. Disaggregating into the agricultural and food and beverage sectors, we find evidence that the negative relationship between GAVCs and income inequality is driven by both participation in the food and beverage as well as the agricultural sectors. Further disaggregating the agriculture and food and beverage sectors into their forward and backward (upstream and downstream) components, we find results supporting our main insights. Both the forward and backward components in agriculture as well as food and beverage sectors are negatively associated with income inequality. These results are robust to instrumental variable estimators using different sets of instruments under different identifying conditions. They are also robust to the GMM approach, where we further establish the negative relationship between the food and beverage sector and income inequality and the positive association between the agriculture sector and income inequality.

¹ The WIID uses a multiple imputation model to compute missing observations which is has some weaknesses (Jenkins 2015).

Delving into some geographical heterogeneities, we further show that in response to greater participation in GAVCs, income inequality decreases in both advanced and emerging markets and developing economies. This is an important result that points to opportunities inherent in GAVCs for reducing inequality in both developed and developing countries. Important to note is that these results are maintained for both the agriculture and the food and beverage sectors. Looking at some additional geographical heterogeneity results, we also find significant contextual differences across Africa, Asia, Europe, and South America. In terms of explaining the negative relationship between GAVCs and income inequality, we consider possible job creation effects, structural transformation, and formality in the economies. We see that participation in the food and beverage sector is associated with reductions in the share of informal output and self-employment. That is, the food and beverage sector reduce informality and to some extent create more formal jobs in the industrial and service sectors. The agricultural sector, in contrast, results in more informal output and self-employment. We also show that structural transformation could explain these results. While participation in the agricultural sector increases employment in agriculture, it reduces employment in the industrial and service sectors. Participation in food and beverage sector, on the other hand, reduces employment in agriculture and pushes up employment in the service and industrial sectors.

We offer several contributions to the growing literature on the implications of GAVCs from a global perspective. First, we establish the income inequality implications of participation in GAVCs. Previous studies have shown that GAVCs are associated with economic performance, growth, and structural transformation but few or no studies have looked at links with income inequality (Carpa and Martínez-Zarzoso 2022). Most of the existing evidence on GAVCs and inequality has mostly been confined to advanced, mostly OECD countries (World Bank, 2020) While it can be assumed that growth and inequality follow a mechanistic relationship, this is not a straightforward relationship; economic advances cannot always be assumed to lead to positive social outcomes (Barrientos et al. 2011). In some cases, growth may lead to greater inequality as the poor try to catch up while the rich get richer

(Bourguignon 2016). In terms of our measures of inequality, we use the Gini indices, which measure inequality within countries.

Second, we highlight significant heterogeneity in the relationship between GAVCs and income inequality. Specifically, we show that the negative relationship between GAVC participation and income inequality is driven by both the agricultural as well as the food and beverage sector. These heterogeneous relationships are further echoed in the geographical disaggregation analysis, where we look at countries by income grouping (advanced economies versus emerging economies) as well as by continent (Africa, Asia, Europe, and Latin America). Our third contribution relates to mapping and documenting GAVC participation from a global perspective. We further disaggregate these value chain groups into forward and backward linkages, representing the different nodes of the various chains. We show how income inequality responds to participation in each of these components of GAVCs.

Fourth, we support and corroborate some important insights related to the growth and structural transformation impacts of agrifood value chains (Lim 2021; Lim and Kim 2022; Montalbano and Nenci 2022). We also show the job creation impacts of GAVC participation, highlighting the divergent role of employment formality (self-employment versus wage employment in both upstream and downstream segments). Lim and Kim (2022) highlighted the positive job creation effects of GAVCs, which they show are largely driven by the downstream food and beverage sector. We support these findings and attempt a marginal view on job quality which we proxy using formality following precedence from the labour literature (Hovhannisyan et al. 2022). The final contribution, which overlaps with the contributions described above, relates to data. We provide evidence from a constructed panel of 167 countries over a 31-year period (1990 to 2020). To our knowledge, we are the first to provide evidence from such a comprehensive recent dataset. Most previous studies rely on shorter panels (up to 2015) sourced from the UNCTAD-Eora Global Value Chain (GVC) database (Balié et al. 2019; Lim 2021; Lim and Kim 2022; Montalbano and Nenci 2022; Nenci et al. 2022).

The rest of this article is structured as follows. After the introduction, section two presents a brief conceptual linkage and literature review. Section three provides the data used for the analysis and some descriptive statistics and insights from the data. In section four, we present the empirical strategy used in examining the relationship between GAVCs and income inequality. The results are then presented and discussed in section five while the manuscript concludes in section six with some summary and policy implications.

2. Conceptual linkage and literature review

To structure our thinking on the relationship between GAVCs and income inequality, we conceptually link them by taking a deep dive into the literature on value chains. Most of the insights here are based on the well-established local value chains literature which is somewhat related with immense implications for global agrifood value chains. (Feyaerts et al. 2020). We also borrow some thoughts from the trade literature given that About 80 percent of global trade occurs through GVCs (UNCTAD 2013). Participation in global production networks has been shown to have significant micro- and macro-level benefits. Particularly, participation in agrifood value chains have been shown to trigger structural transformation through increases in growth and changes in the shares of employment in agriculture and service sectors (Lim 2021). This growth has also been accompanied with growth in labor market and value added per worker (Montalbano and Nenci 2022; Lim and Kim 2022). These job creation impacts have been driven by the downstream food and beverage processing sector as opposed to the upstream agriculture sector.

Despite the positive economic and growth implications of GVCs participation, there may be inequality concerns given that the gains from growth may not be homogenous (Ravallion 2001; Antràs et al. 2017). That is, the poor may not share the gains from this rising growth as well as the losses from any possible contraction. The poor may also bear the brunt of any negative implications of GVCs given their prevalent vulnerability. But this is inconclusive as participation in high-value export chains (fruits, vegetables, and cut flower) has been associated with significant wage employment effects in developing

countries, especially in rural areas, which are associated with income increases. This impact seems to be strongest for the poorest households with the ability to reduce inequality (Maertens and Swinnen 2009; Schuster and Maertens 2013; van den Broeck et al. 2017).

However, rural wage employment could also be associated with increased inequality, given that the poor are usually excluded from these high-value and profitable value chains, and are often subject to low wages and poor working conditions (Barrientos et al. 2011; Ortiz and Aparicio 2007). But beyond these micro-level impacts, and with insights from the pioneering work of Kuznets and Lewis, there could be an important, though mechanistic relationship between growth and income inequality (Anand and Kanbur 1993). This hypothesis is based on the pessimistic conjecture that inequality rises at the early stages of growth and development (Lundberg and Squire 2003). GVCs could increase and exacerbate inequality but they also have the potential to reduce and mitigate it (Grabs and Ponte 2019; Lang et al. 2022).

From this, one would expect an ambiguous relationship between GAVCs and income inequality and this could occur through different pathways but mainly income and employment. Reduction in income inequality could arise from some of the opportunities and benefits inherent in GAVCs such as income increases (van den Broeck et al. 2017) which could lead to investments in assets and livestock (Scoones et al. 2018) which represents aspects of rural wealth and prosperity (Tabe Ojong et al. 2022). All these have the potential to broaden agricultural production and scope through further reinvestments into the sector. For instance, the development of various crop value chains in some countries especially in West Africa have been associated with some aspects of processing, technology adoption and infrastructural development (Yaro et al. 2017).

Beyond this, there could be some other aspects of economic development captured through job creation effects and value-added per worker (Lim and Kim 2022; Montalbano and Nenci 2022). There exist immense employment opportunities in many agrifood value chains be in the agricultural or food

and beverage sub-sectors. Some examples include the horticultural chains in Namibia, Kenya, Senegal and Egypt, the tobacco value chains in Southern African countries such as Malawi and Zimbabwe, cocoa value chain in West Africa, sugar value chain in Brazil and the oil palm value chain in Southeast Asia (Maertens and Swinnen 2009; van den Broeck et al. 2017; Scoones et al. 2018; Feyaerts et al. 2020; Tabe-Ojong and Abay 2023). For most of these upstream agricultural sectors, self-employment is usually key while the downstream food and beverage sector usually involves gainful employment. This differentiation may induce aspects of informality where the agricultural sub-sector may be associated with informality and the food and beverage sector with formality, although this is very much arguable. This may further suggest that the food and beverage downstream sector is usually more regulated than the traditional agricultural sector. Given the less regulation that is usually common place with informality and self-employment, there could be issues of poor working conditions, social issues and formality (Barrientos et al. 2011). Moreover, formality is increasingly being used as a proxy for job quality which is lower in many developing countries (Hovhannisyan et al. 2022)

3. Data and Descriptive Statistics

To understand the relationship between GAVCs and income inequality, we rely on different datasets. For the GAVCs, we constructed a panel dataset of 167 countries over a 31-year period (1990–2020). We used the input-output (IO) tables from EORA-MRIOs tables sourced from the UNCTAD-Eora Global Value Chain database. For our outcome measure, we collate information on income inequality from the World Income Inequality Database (WIID)² hosted by UNU-WIDER. For our control variables which are grouped under agriculture, socio-economic, demographic and trade/policy or institutional constraints, we rely on the world development indicators database of the world bank where we obtain all these variables.

3.1 Global Agrifood Value Chain participation

² The WIID uses a multiple imputation model to compute missing observations, which has some weaknesses (Jenkins 2015).

There have been several attempts to accurately measure GVC participation (Koopman et al. 2014; Wang et al. 2017; Los and Timmer 2018; Belotti et al. 2021). For our analysis, we employ the EORA-MRIOs, generated by the UNCTAD-Eora Global Value Chain Database to measure GAVC participation, following the framework proposed by Borin and Mancini (2019).³ The framework captures all the sources of value added activities for more than two countries, which are often missing in other measures of GVCs, and has recently been used by several empirical papers (Borin et al. 2021; Lim and Kim 2022).

Following this extensive literature on GAVCs, we decompose gross agrifood exports into broad value-added activities, allowing us to estimate GAVC participation. Gross agrifood exports are decomposed into several components including domestic agrifood value-added exports (DVX_{it}), and foreign agrifood value added (FVA_{it}). DVX_{it} refers to the value of domestic agrifood products exported and used by third countries as intermediate goods for their exports and re-exported. FVA_{it} refers to the value of agrifood products exports that originates from imported inputs. Follow Borin and Mancini (2019) GAVC participation is measured as follows:

$$GAVC_{it} = \frac{DVX_{it}^{Total\ agri} + FVA_{it}^{Total\ agri}}{Gross\ Export_{it}^{Total\ agri}} \quad (1)$$

Where $\frac{DVX_{it}^{Total\ agri}}{Gross\ Export_{it}^{Total\ agri}}$ and $\frac{FVA_{it}^{Total\ agri}}{Gross\ Export_{it}^{Total\ agri}}$ represents respectively forward GAVC participation in country I at period t and backward GAVC participation in country i at period t . Total agriculture ($Total\ agri$) refers to both agriculture and food and beverage. Thus equation (1) can be expressed as follows:

$$GAVC_{it} = \frac{DVX_{it}^{agri} + FVA_{it}^{agri} + DVX_{it}^{FB} + FVA_{it}^{FB}}{Gross\ Export_{it}^{agri} + Gross\ Export_{it}^{FB}} \quad (2)$$

³ For similar analytical frameworks that have been developed to measure supply and demand contributions of countries and sectors in GVCs, see Koopman et al. 2014, Wang et al. 2017, and Los and Timmer 2018.

Calculating sectorial level of GAVC participation follows the same process. Thus, to calculate GAVC participation, we use the agricultural sector classification to measure agricultural GVCs (GaVC) and the food and beverage sector classifications to measure food GVCs (GfVC), respectively as follow:

$$GaVC_{it} = \frac{GVC_{it}^{agri}}{Gross\ Export_{it}^{agri}} = \frac{DVX_{it}^{agri} + FVA_{it}^{agri}}{Gross\ Export_{it}^{agri}} \quad (3)$$

$$GfVC_{it} = \frac{GVC_{it}^{FB}}{Gross\ Export_{it}^{FB}} = \frac{DVX_{it}^{FB} + FVA_{it}^{FB}}{Gross\ Export_{it}^{FB}} \quad (4)$$

Following Belotti et al. (2021) and using the general cross-country IO table from the UNCTAD-Eora GVC Database, we measure country-level GAVC participation for 167 countries in the period 1990–2020.

Table 1: Summary Statistics: GAVC Participation (1990–2020, N=167 countries)

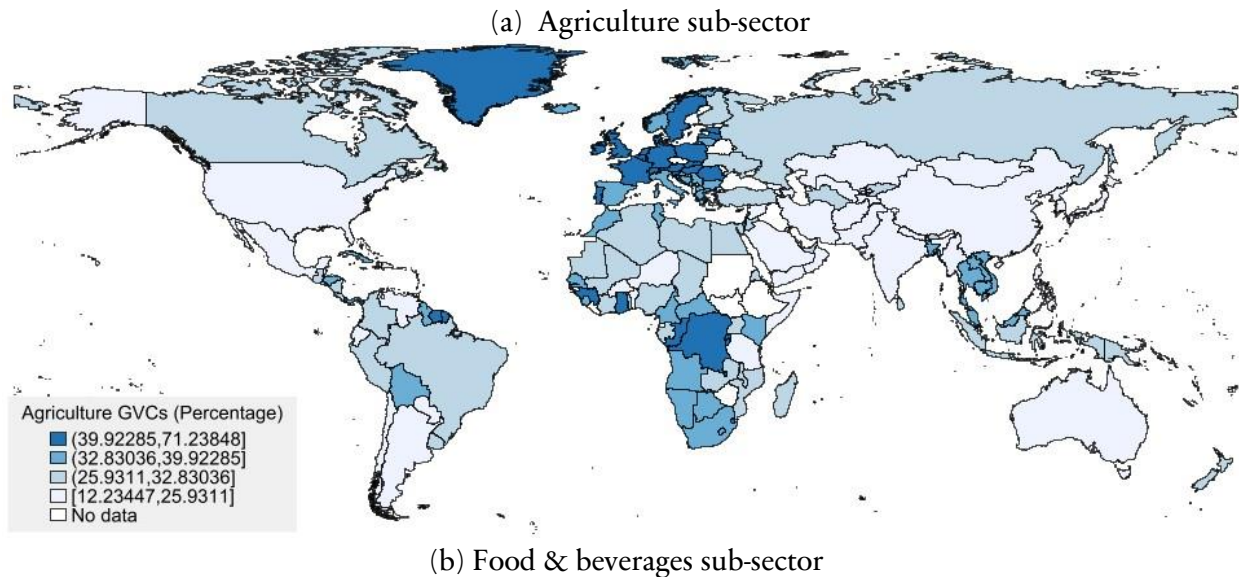
Variable	Obs	Mean	Std. Dev.	Min	Max
<i>GAVC participation</i>					
GAVC (%)	5177	32.653	11.216	0.863	81.089
Forward Participation (%)	5177	16.83	8.35	0.739	53.671
Backward Participation (%)	5177	15.823	11.209	0.037	69.113
<i>Agriculture GVC participation</i>					
Agriculture GVC (%)	5177	33.565	11.641	0.799	82.861
Forward Participation (%)	5177	22.617	8.971	0.684	67.814
Backward Participation (%)	5177	10.948	9.015	0.021	78.762
<i>Food and Beverage GVC participation</i>					
Food and beverage GVC (%)	5177	32.936	12.131	0.892	88.256
Forward Participation (%)	5177	12.669	7.112	0.763	48.699
Backward Participation (%)	5177	20.267	11.941	0.101	81.035

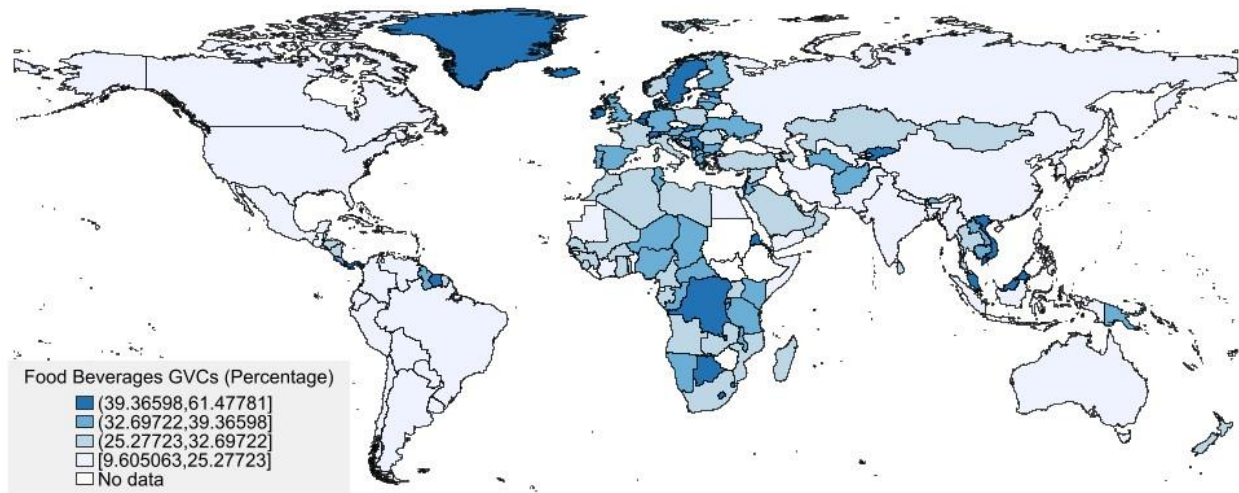
Note: Authors' calculations based on data sourced from the UNCTAD-Eora Global Value Chain (GVC) database and following Belotti, Borin and Mancini (2020). GVC participation is measured by a GVC share of a country's gross exports following Koopman, Wang, and Wei (2014). Backward participation is measured by the foreign value added (FVA); forward participation is measured by the domestic value added (DVX). Agrifood includes both the agricultural industry and the food and beverage industry.

Table 1 shows descriptive statistics of GAVC participation for 167 countries for the period 1990–2020. Across the sample countries, the mean GAVC participation is 32.7%. The decomposition of the GAVC

by -sectors (agriculture vs. food and beverage) shows that agricultural GVC participation (33.6%) is slightly larger than food GVC participation (32.9%). For each sector, GVC participation can be divided into forward (upstream) and backward (downstream) GVC participation. While the GAVC participation is almost equally distributed between forward GVC participation (16.8%) and backward GVC participation (15.8%), agricultural and food and beverage sub-sectorial decompositions are different. For the agricultural sector, forward GVC participation is on average (22.6%) more than double backward GVC participation (10.9%). The situation is different and somewhat opposite for the food and beverage sector, for which forward GVC participation is less than backward GVC participation (12.7% and 20.3% respectively). These results follow previous findings and suggest that the food and beverage sector accounts for a larger share of backward linkages in production while the agricultural sector accounts more of forward linkages. This is intuitive and can be explained by the fact that the food and beverage sector involves a higher degree of foreign value added embodied in industrialized/semi-industrialized products (Lim and Kim 2022).

Figure 1: GAVC participation (percentage) across countries (average 1990–2020)



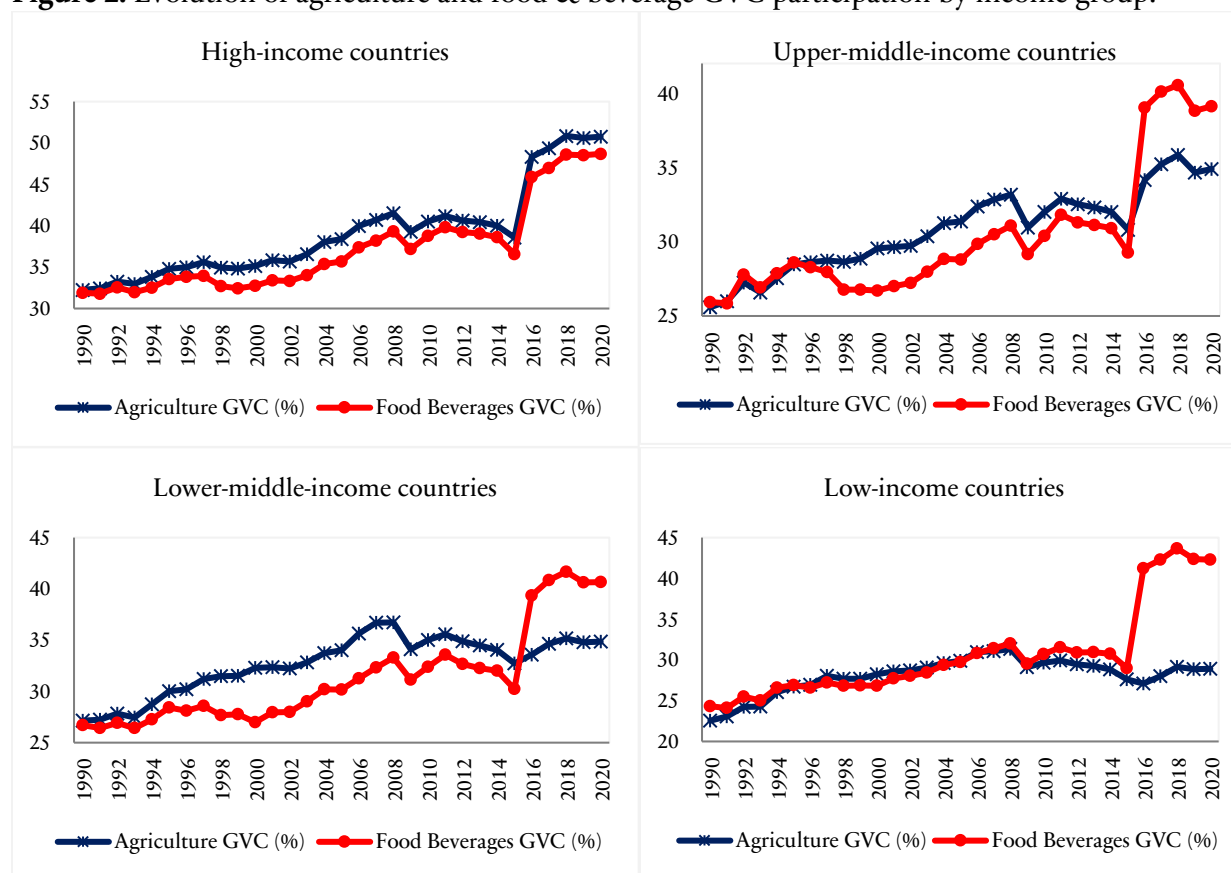


Source: Authors' calculations based on EORA-MRIO data.

Country-level representation of the average agricultural and food and beverage sectors show differences across countries. Figure 1 shows the geographic distribution of the average agricultural and food and beverage sectors during the period 1990–2020. Europe and sub-Saharan Africa (SSA) show a relatively high-level of GVC participation in both the agriculture and food sectors when we look at GVC as a share of total exports. However, considering the level of agricultural GVC participation in US dollars, sub-Saharan Africa is lagging behind America Europe and Asia. The high level of agricultural GVC participation in percentage of gross exports in sub-Saharan Africa can be explained by increased demand for commodities and agricultural raw materials from advanced economies (Balié et al. 2019; Feyaerts et al. 2020).

Figure 2 shows the trends in both the agricultural and food and beverage sub-sectors from 1990 to 2020 by income groups. High-income countries show the highest level of GAVC participation in the agriculture sub-sector in both 1990 and 2020. The figures indicate a significant increase in the level of agricultural GVC participation in 2016 for all income groups, especially in food and beverage GVC participation.

Figure 2: Evolution of agriculture and food & beverage GVC participation by income group.



Source: Authors' calculations based on EORA-MRIO data.

3.2 Income Inequality

We use the standard Gini coefficient to proxy for income inequality. This is obtained from the World Income Inequality Database (WIID)⁴ hosted by UNU-WIDER. It ranges between 0 and 100 and has been widely used as a measure of inequality within countries. Table 2 shows the summary statistics of the income inequality measures. It also shows the summary statistics of the pathways outcomes such as self-employment, informal output, employment in agriculture industry and the service sectors. All these variables are obtained from the world development indicators database. The table shows that inequality is relatively higher in the sample. With a level of inequality of about 682 measured by the

⁴ The WIID uses a multiple imputation model to compute missing observations which has some weaknesses (Jenkins 2015).

absolute Gini coefficient⁵ (43.6 percent for the standardized Gini coefficient), the current sample highlights an important level of informality (32 percent). Moreover, the services sectors employ most of the workforce (50 percent), followed by the agricultural sector. This ranking of the agricultural sector as the second-best employer of the population makes it important to study the GAVCs' participation.

Table 2: Summary Statistics of outcome variables

Variable	Obs	Mean	Std. Dev.	Min	Max
<i>Gini coefficients</i>					
Gini coefficient	5177	43.646	11.463	0	77.085
Absolute Gini	5177	682.282	685.272	0	4452.831
<i>Informality</i>					
Self-Employment	2478	31.622	21.255	0.396	95.5
Informal output	4110	31.439	12.14	7.967	67.664
<i>Structural Transformation</i>					
Employment Agriculture (%)	4640	29.658	24.408	0.03	92.37
Employment Industry (%)	4640	20.112	8.702	0.28	59.58
Employment Services (%)	4640	50.231	18.735	5.34	88.75

Source: Data from the World Income Inequality Database (WIID) hosted by UNU-WIDER and WDI.

4. Estimation Strategy

4.1 Baseline Regression Model

To establish the relationship between GAVC participation and income inequality and given the nature of our dataset, we delve directly into some basic panel data analysis. Our first empirical strategy is based on the standard linear regression methods with country and time fixed effects. Our empirical model is specified as:

$$Gini_{i,t} = \delta + \eta_i + \varphi_t + \alpha GAVC_{i,t} + \sum_{j=1}^p \beta_j X_{i,t}^j + \varepsilon_{i,t} \quad (5)$$

where η_i and φ_t represent country-specific and time-specific effects respectively. Gini represents absolute Gini coefficient, which measures the level of income inequality within a country; $GAVC_{i,t}$ represents three measures of GAVC participation (% of gross agriculture exports) – agricultural sector

⁵ The absolute Gini coefficient has a larger range because absolute levels of inequality are calculated as the relative Gini multiplied by the country's GDP (UNU-WIDER 2023).

participation (As), food beverage sector participation (FBs) and the combined GAVCs participation (GAVC). We include a battery of controls to improve the precision of the model. Our parameter of interest is α which specifies the relationship between GAVCs and income inequality. $X_{i,t}^j$ represents a set of control variables which can broadly be grouped into agriculture, socio-economic, demographic and trade policy and institutional factors. We control for socio economic factors such as gross domestic production (GDP) per capita Growth, Foreign Direct Investment (in percentage of GDP), export of goods and services (in percentage of GDP), import of goods and services (in percentage of GDP), school enrollment ratio (in percentage of gross enrollment), level of unemployment (in percentage of total labor force), and inflation/consumer price index. We also consider agricultural factors measured by land areas and cereals production. Additionally, we include demography/population through the following variables: population aged between 0 and 14 (in percentage of total population), population aged between 15 and 64 (in percentage of total population), population of females (in percentage of total population), population of males (in percentage of total population), population density, rural population (in percentage of total population) and urban population (in percentage of total population). Finally, we consider trade policy, including Regional Trade Agreement, Free Trade Agreement, and Currency Union as control variables.

4.2 Identification

Estimating Equation (5) is not void of some empirical challenges notably endogeneity concerns which could arise from reverse causality, unobserved heterogeneity, and measurement error. Given that the level of GAVC participation of a given country is not a random process and given that participation in GAVCs is likely not exogenous to inequality, it is important to discuss these potential threats to identification and how they matter for the analysis. For reverse causality, while one may expect that growth in GAVC participation may reduce inequality, participation in GAVCs may also result from these outcomes. For unobserved heterogeneity, there may be unobserved country level differences driving participation in GAVCs and inequality and these may even be determined by the same process. Finally, for measurement error, it is hard to claim the accuracy of the data generating process to

measurement error and we acknowledge that this is admittedly not perfect. Given these threats to identification, we further employ some additional identification strategies besides the OLS with country fixed effects regression. We use the two staged least squares approach where we employ the Bartik shift share IV strategy. We also robustify this somewhat robust IV with some other external instruments in the standard 2SLS approach. We also use consider dynamic panel models such as the Generalized Method of Moments (GMM) approach.

4.2.1 The Bartik Shift-share instruments

Following Lim and Kim (2022), we rely on shift-share instruments developed by Bartik (1991) to capture the potential exogenous variation in countries' GAVC participation. This IV approach allows the reduction of potential endogeneity sources, including reverse causality and omitted variables (Card 2009). The principle of this IV approach is to decompose the endogenous explanatory variable, in our case GAVCs into an accounting identity with multiple component parts. By decomposing this into subdimensions, Bartik shift-share IVs predict the treatment variation by the subdimension-specific share at a given point in time (a share) and the overall variation across time in a subdimension-specific variable (a shift) (Goldsmith-Pinkham et al. 2020; Lim and Kim 2022).⁶ Our subdimensions are the agricultural industries: (i) the agricultural sector and (ii) the food and beverages sector. Given this, we instrument the GAVC participation with the following Bartik instrument:

$$IV_{i,t} = \frac{1}{Gross\ Exp_{i,t}} (W_{i,t-1}^{Agri} e_t^{Agri} + W_{i,t-1}^{FB} e_t^{FB}) \quad (6)$$

With

$$W_{i,t-1}^{Agri} = \frac{GVC_{i,t-1}^{Agri}}{\sum_i GVC_{i,t-1}^{Agri}}; W_{i,t-1}^{FB} = \frac{GVC_{i,t-1}^{FB}}{\sum_i GVC_{i,t-1}^{FB}}; e_t^{Agri} = \sum_{i=1}^n GVC_{i,t}^{Agri} \text{ and } e_t^{FB} = \sum_{i=1}^n GVC_{i,t}^{FB}$$

⁶ For example, Bartik 1991 instruments the local employment growth rate by local industry employment shares and country level industry employment growth rates.

$AgriGVC_{i,t}$ represents agricultural GVCs participation, $FBGVC_{i,t}$ is food and beverage GVCs participation; $Gross\ Exp_{it}$ represents gross exports, and $n=167$ is the number of countries.

4.2.2 External instruments

Beyond the Bartik shift share instrument, we also rely on other instruments such as climate-related variables including change in temperature and climate altering land cover index and the availability of arable lands. All the instruments are relevant and can be argued to be exogenous to income inequality especially given that they are weather related and based on geographical endowments. In fact, changes in temperature result from climate change that has been accelerated by human activities. The availability of arable land is geographical and can be argued to be regionally based, whose presence is mostly due to natural endowments. Some regions possess more arable land than others and this is an exogenous phenomenon because it is not the result of any other economic activity. Changes in temperatures as well as agricultural lands are directly linked to agricultural production and trade. Moreover, the only way changes in temperatures and agricultural lands can impact income inequality is through agricultural production in a broader sense. We are not the first to use these kinds of IVs as they have been used in some empirical analysis in the bigger trade literature. For instance, Brueckner and Lederman (2015) for their IV strategy used rainfall as an instrument for trade openness in Africa (SSA); a strategy that builds on earlier work (Miguel et al. 2004; Barrios et al. 2010). However, given that innovative irrigation methods are now commonplace, we relied on changes in temperatures that are difficult to regulate and depend on natural phenomena. All the instruments meet the IV relevance condition and as we have argued, they may also be exogenous, making them possibly valid IVs. Beyond conceptually arguing about IV exogeneity, there are usually no valid tests for this. But given that we have three IVs, we can infer on instrument validity by performing the Sargan-Hansen test of overidentifying restrictions. We fail to reject the (joint) null hypothesis that these IVs are valid.

4.3 The Generalized Method of Moment: Internal instruments – lags of explanatory variables

As highlighted earlier, we also use the systems GMM approach. GMM is another useful technique that can deal with the problem of endogeneity. Originally developed by Holtz-Eakin et al. (1988), it was

advanced by Arellano and Bond (1991). The GMM approach can be summarized in three steps: (1) specify the regression equation as a dynamic panel data model, (2) take first differences to remove unobserved time-invariant country-specific fixed effects, and (3) instrument the right-hand-side variables in the first-differenced equation using levels of the series lagged two periods or more while ensuring that the model is not overidentified (for details see Blundell and Bond (1998). The model is specified as follows:

$$Gini_{i,t} = \delta + \eta_i + \varphi_t + \phi Gini_{i,t-1} + \alpha GAVC_{i,t} + \sum_{j=1}^k \beta_j X_{i,t}^j + \varepsilon_{i,t} \quad (7)$$

where η_i and φ_t represent country-specific and time-specific effects respectively. Gini represents absolute Gini coefficient, which measures the level of income inequality within a country; $GAVC_{i,t}$, represents GAVC participation and $X_{i,t}^j$ represents a set of control variables for i countries over time t , and h is the number of control variables where $h = 1, 2, \dots k$.

5. Results and Discussion

5.1 Baseline estimates

We begin by highlighting some insights from a simple scatter plot which shows a quasi-linear relationship between GAVCs and income inequality. Figure 3A shows a somewhat inverted U-shaped relationship between the agriculture sub-sector and the Gini coefficient (a positive relationship followed by a negative relationship). Figure 3B shows a quasi-linear relationship between food and beverage sub-sector and inequality.

[illegible]

(1) (2) (3) (4)

Economy	Yes	Yes	Yes	Yes
Agriculture	Yes	Yes	Yes	Yes
Population	Yes	Yes	Yes	Yes
Demography	Yes	Yes	Yes	Yes
Trade Policy	Yes	Yes	Yes	Yes
Observations	2,836	2,836	2,836	2,836
R-squared	0.634	0.629	0.634	0.634
Number of countries	144	144	144	144
Country FE	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes

Note: All estimations include controls that are grouped under economy, agriculture, population, demography, and trade policy. They include Gross Domestic Production (GDP) per capita Growth, Foreign Direct Investment (in percentage of GDP), export of goods and services (in percentage of GDP), import of goods and services (in percentage of GDP), school enrollment ratio (in percentage of gross enrollment), the level of unemployment (in percentage of total labor force), inflation/consumer price index, land areas, cereals production, population ages between 0 and 14 (in percentage of total population), population aged between 15 and 64 (in percentage of total population), population of females (in percentage of total population), population of males (in percentage of total population), population density, rural population (in percentage of total population), urban population (in percentage of total population), regional trade agreement, free trade agreement, and currency union. Standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

We now move to aspects of GAVC position by exploring forward and backward linkages in both the agriculture and food sectors. Forward GVC participation refers to the domestic value added in intermediate goods that are further re-exported by the partner country, while backward GVC participation is foreign value added embodied in gross exports. The results of the baseline estimates (Table 4) suggest a negative relationship between both forward and backward GAVCs and income inequality. However, forward GAVC participation has a higher absolute coefficient, suggesting that forward GAVCs may yield greater associational gains in reducing income inequality (Table 4, column 3). For the agriculture sub-sector, we only observe a negative relationship between forward agriculture GVCs and income inequality while for the food and beverage sub-sector, we obtain negative associations for both the forward and backward linkages like in the case of the GAVCs.

Table 4: GAVC position and Income inequality

	(1) Gini	(2) Gini	(3) Gini
Forward GAVC (%)	-5.821***		

	(1.009)		
Backward GAVC (%)	-3.285***		
	(0.546)		
Forward agricultural sector (%)	-4.752***		
	(0.792)		
Backward agricultural sector (%)	-0.132		
	(0.569)		
Forward food sector (%)		-4.805***	
		(1.046)	
Backward food sector (%)		-2.648***	
		(0.424)	

Economy	Yes	Yes	Yes
Agriculture	Yes	Yes	Yes
Population	Yes	Yes	Yes
Demography	Yes	Yes	Yes
Trade Policy	Yes	Yes	Yes
Observations	2,836	2,836	2,836
R-squared	0.635	0.634	0.635
Number of countries	144	144	144
Country FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes

Note: All estimations include controls that are grouped under economy, agriculture, population, demography, and trade policy. They include Gross Domestic Production (GDP) per capita Growth, Foreign Direct Investment (in percentage of GDP), export of goods and services (in percentage of GDP), import of goods and services (in percentage of GDP), school enrollment ratio (in percentage of gross enrollment), the level of unemployment (in percentage of total labor force), inflation/consumer price index, land areas, cereals production, population ages between 0 and 14 (in percentage of total population), population aged between 15 and 64 (in percentage of total population), population of females (in percentage of total population), population of males (in percentage of total population), population density, rural population (in percentage of total population), urban population (in percentage of total population), regional trade agreement, free trade agreement, and currency union. Standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

5.2 GAVC and inequality

5.2.1 IV estimates of GAVCs and inequality

We now present the results of the IV approach where we show insights from the main IV approach where we rely on two different IV approaches. The first type relies on a set of instruments including temperature, climate altering land cover index, and land availability. The second type of instrument relies on the shift-share instruments approach developed by Bartik (1991). The IV specifications satisfy

the requirements of diagnostic tests. The Hansen J-test is applied to test the null that the instruments are jointly valid. The p-values for the tests in all specifications are high (higher than 0.1). The conclusion is that the overidentifying restrictions are not rejected (See Table A 30 in the Appendix). In the case of the shift-share instrument, it is exactly identified because it relies on one instrument. In addition, the weak identification test was performed on the shift share instrument to test for the strength of the instrument. The Cragg-Donald Wald F statistic (F statistic = 39.335) is higher than Stock-Yogo critical values. The results indicate that the instruments are not weak (see Table A 30 in the Appendix).

Using these instruments, we obtain results that are similar and in the same direction with the country fixed effects OLS regressions in terms of magnitudes, size, and direction. Of course, OLS and IV estimations are different since they have different assumptions and estimate different effects. While OLS gives average treatment effects, IV estimators produce local average treatment effects. As shown in Table 5, GAVC is negatively associated with income inequality. This is mostly driven by participation in the downstream food and beverage sector. Although we find a negative relationship with the agricultural sector, this is not statistically different from zero. The results of the IV approach using the shift-share instrument further confirm the baseline country FE OLS regressions as well as the IV estimates. In fact, the Bartik shift share instrumental variable approach also highlights a negative association between GAVC participation and income inequality. To check the robustness of our IV approach, we estimated alternative models following Goldsmith-Pinkham et al. (2020). We relied on alternative estimators by estimating by limited information maximum likelihood (Anderson and Rubin, 1949), by modification of the bias-corrected two-stage least square (MBTSL; see Kolesar et al., 2015), and the information matrix-based standard errors (IM-SE) as a first check of the specification of our models (Table A25). For all estimators, the coefficients and standard errors are like our core results, providing no reason to assume that the model is misspecified. In addition to this approach, we also estimate a model with individual industry shares as separate IVs to assess the exogeneity of industry shares. The findings highlight a non-significant p value confirming the exogeneity of our share-driven

instruments. All these bolsters and strengthen the insight that GAVCs can reduce income inequality. The potentials of agri-food GVCs in reducing poverty among smallholder farmers in developing countries have been acknowledged (World Bank 2019; FAO 2020). In this vein GAVCs could be associated with lower income inequality and the drive towards shared prosperity by improving the comparative advantage of countries. Smallholder farmers in many developing countries can access agri-food markets which may lead to reduction in food prices (Dalheimer et al. 2023).

Table 5: Results – IV approach - Agri GVC participation and income inequality

	IV approach				Shift-Share IV
	(1) Gini	(2) Gini	(3) Gini	(4) Gini	(5) Gini
GAVC (%)	-10.53** (5.167)				-8.200* (4.504)
Agriculture (%)		-8.501* (5.102)		1.591 (10.08)	
Food and beverage (%)			-7.905** (3.865)	-8.960 (7.732)	
Economy	Yes	Yes	Yes	Yes	Yes
Agriculture	Yes	Yes	Yes	Yes	Yes
Population	Yes	Yes	Yes	Yes	Yes
Demography	Yes	Yes	Yes	Yes	Yes
Trade Policy	Yes	Yes	Yes	Yes	Yes
Observations	2,609	2,609	2,609	2,609	2,836
Number of countries	141	141	141	141	144
Country FE	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes

Note: All estimations include controls that are grouped under economy, agriculture, population, demography, and trade policy. They include Gross Domestic Production (GDP) per capita Growth, Foreign Direct Investment (in percentage of GDP), export of goods and services (in percentage of GDP), import of goods and services (in percentage of GDP), school enrollment ratio (in percentage of gross enrollment), the level of unemployment (in percentage of total labor force), inflation/consumer price index, land areas, cereals production, population ages between 0 and 14 (in percentage of total population), population aged between 15 and 64 (in percentage of total population), population of females (in percentage of total population), population of males (in percentage of total population), population density, rural population (in percentage of total population), urban population (in percentage of total population), regional trade agreement, free trade agreement,

and currency union. Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

5.2.2 GMM estimates of GAVCs and inequality.

As earlier mentioned, we also rely on the GMM approach to address endogeneity concerns,⁷ but also to provide more robust estimates. Table 6 presents the results of the GMM approach. In each specification, the number of instruments is lower than the number of countries. The Sargan-Hansen test, which provides a test of overidentifying restrictions, shows that the additional instruments associated with the system GMM estimator are valid. The two tests for serial correlation show evidence of first order autocorrelation since the null hypothesis of no autocorrelation is rejected using the AR (1) test. However, we fail to reject the null hypothesis using the AR (1) test. This provides evidence of no second order autocorrelation (See Table 6). Therefore, the results of the diagnostic tests indicate that the GMM estimates are valid. The results also follow both the baseline and the two IV approaches. we again obtain a negative association between GAVC participation and income inequality. This, as we earlier observed, is driven by both the upstream agricultural sector and the downstream food and beverage sector. Including the agriculture and food and beverage sectors together in the same equation, confirms the initial results: both sectors are associated with reductions income inequality (Table 6, column 4).

⁷ The validity of the results of the GMM estimator can be affected by overidentification of instruments and second-order autocorrelation of the residuals. The system GMM estimator performs using lagged values of endogenous variables as instruments. Thus, the number of instruments can easily explode as the time-series dimension of the dataset becomes longer, weakening the test for over-identifying restrictions due to instrument proliferation or overidentification (Roodman 2009). According to Windmeijer (2005), GMM becomes more efficient when the lag length is controlled to use fewer instruments in the estimation. The Sargan or Hansen tests are diagnostic tests designed to establish whether additional instruments associated with the system GMM estimator are valid.⁷ Practically, when using the system GMM estimator, the instrument count can be reduced creating one instrument per variable instead of different instruments for each variable and for each period. The result is a smaller set of instruments without compromising the optimal number of lags required (Roodman 2009). Regarding second-order autocorrelation of the residuals (serially correlated residuals) undermines the results of the GMM estimator by rendering instruments inconsistent. The Arellano–Bond test for first order (AR1) and second order (AR2) serial correlation are used to detect this problem.

Table 6: Results – GMM approach - Agri GVC participation and income inequality

	(1)	(2)	(3)	(4)
	Gini	Gini	Gini	Gini
GAVC (%)	-6.334*** (0.915)			
Agriculture (%)		-4.861*** (0.751)		-2.660*** (0.994)
Food and beverage (%)			-4.388*** (0.733)	-3.640*** (0.688)
Gini Growth _{t-1}	-7.377 (16.54)	60.49*** (15.55)	0.658 (16.31)	15.35 (11.57)
Economy	Yes	Yes	Yes	Yes
Agriculture	Yes	Yes	Yes	Yes
Population	Yes	Yes	Yes	Yes
Demography	Yes	Yes	Yes	Yes
Trade Policy	Yes	Yes	Yes	Yes
Observations	485	485	485	485
Number of countries	141	141	141	141
Number of Inst	114	114	114	126
# Instruments < # countries	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes
AR (1) p-value	0.093	0.230	0.028	0.135
AR (2) p-value	0.553	0.860	0.283	0.453
Hansen p-value	0.326	0.407	0.274	0.261

Note: All estimations include controls that are grouped under economy, agriculture, population, demography, and trade policy. They include Gross Domestic Production (GDP) per capita Growth, Foreign Direct Investment (in percentage of GDP), export of goods and services (in percentage of GDP), import of goods and services (in percentage of GDP), school enrollment ratio (in percentage of gross enrollment), the level of unemployment (in percentage of total labor force), inflation/consumer price index, land areas, cereals production, population ages between 0 and 14 (in percentage of total population), population aged between 15 and 64 (in percentage of total population), population of females (in percentage of total population), population of males (in percentage of total population), population density, rural population (in percentage of total population), urban population (in percentage of total population), regional trade agreement, free trade agreement, and currency union. Standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

The robust confirmation of the positive association between GAVCs and lower inequality could arise from some of the opportunities and benefits inherent in GAVCs such as income increases (van den Broeck et al. 2017) which could lead to investments in assets and livestock (Scoones et al. 2018) which

represents aspects of rural wealth and prosperity (Tabe Ojong et al. 2022). All these have the potential to broaden agricultural production and scope through further reinvestments into the sector. For instance, the development of various crop value chains in some countries especially in West Africa have been associated with some aspects of processing, technology adoption and infrastructural development (Yaro et al. 2017). Beyond this, there could be some other aspects of economic development captured through job creation effects and value-added per worker (Lim and Kim 2022; Montalbano and Nenci 2022).

5.3 Income group and regional heterogeneity

After understanding the association between GAVCs and income inequality, we also examine underlying heterogeneity by looking at income but also geographic heterogeneity. We want to investigate whether the impact of GAVCs varies depending on countries' income characteristics and regional factors. For income heterogeneity, we perform some disaggregated analysis looking at a subsample of advanced economies (AEs) and emerging market and developing economies (EMDEs). Table 7 summarizes the findings of this analysis. We show that GAVC participation is negatively associated with income inequality for both advanced and emerging markets and developing economies. In terms of the sectors, we obtain some heterogeneity where we show that the food and beverage sector is associated with lower inequalities in advanced countries. On the other hand, the agriculture sector is positively associated with inequality in these countries. For emerging markets and developing economies, we observe a negative association for both the agriculture as well as food and beverage sectors with income inequality. We also underscore substantial heterogeneity when we look at the World Bank income classifications: high income countries, upper- and middle-income countries, lower- and middle-income countries and low-income countries (Tables A7-A10 in the supplementary materials).

Table 7: Results – agriculture GVC participation and Income inequality by income group

	AEs				EMDEs			
	(1) Gini	(2) Gini	(3) Gini	(4) Gini	(5) Gini	(6) Gini	(7) Gini	(8) Gini
GAVC (%)	-4.894*** (1.196)				-4.090*** (0.486)			
Agriculture (%)		1.423 (1.238)		8.159*** (1.538)		-2.665*** (0.474)		-1.708*** (0.526)
Food (%)			-5.156*** (1.114)	-9.812*** (1.402)			-2.353*** (0.381)	-1.747*** (0.423)
Economy	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Agriculture	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Population	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Demography	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Trade Policy	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	812	812	812	812	2,024	2,024	2,024	2,024
R-squared	0.849	0.845	0.850	0.855	0.596	0.587	0.589	0.591
Number of countries	32	32	32	32	112	112	112	112
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Note: All estimations include controls that are grouped under economy, agriculture, population, demography, and trade policy. They include Gross Domestic Production (GDP) per capita Growth, Foreign Direct Investment (in percentage of GDP), export of goods and services (in percentage of GDP), import of goods and services (in percentage of GDP), school enrollment ratio (in percentage of gross enrollment), the level of unemployment (in percentage of total labor force), inflation/consumer price index, land areas, cereals production, population ages between 0 and 14 (in percentage of total population), population aged between 15 and 64 (in percentage of total population), population of females (in percentage of total population), population of males (in percentage of total population), population density, rural population (in percentage of total population), urban population (in percentage of total population), regional trade agreement, free trade agreement, and currency union. Standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Beyond these income classifications, we also undertake some geographical heterogeneity where we look at different continental regions including Africa, Asia, Europe, and South America. We again observe considerable heterogeneity between the continents. For African countries, GAVC is positively associated with income inequality (Table A1, column 3). This positive association is driven by the agricultural sector that has a positive and significant coefficient (Table A1, columns 1 and 4). However, food and beverage GVC participation is negatively associated with income inequality (Table A1, columns 2 and 4). In Asia, GAVCs are negatively associated with income inequality, and this is mostly driven by the food and beverage sector. The case of Europe seems like a mix of both Africa and Asia as we observe a negative association between GAVCs and income inequality. There are some differences

between the agriculture and food and beverage sector: while the food and beverage sector is negatively associated with income inequality, the agricultural sector is positively associated with income inequality. Finally, for South America, we do not observe any statistical relationship between GAVCs and income inequality. However, further looking at the sectors, we observe a positive association between the food and beverage sector and income inequality. All these findings show the existence of substantial heterogeneity and possible masking of effects which get hidden in an aggregated analysis. The positive association between agrifood value chains and income inequality in Africa and South America corroborates some evidence on trade and earnings inequality in Ecuador (Adao et al., 2022)

5.4 Pathways – structural change and informality

After establishing the relationship between GAVCs and income inequality as well as digging into some heterogeneity, we explore what could be driving these relationships by exploring two factors: job creation and informality and structural transformation.

5.4.1 GAVCs, job creation, and informality

Table 8 presents the relationship between GAVC participation and self-employment as well as informal output.⁸ Relying on a panel fixed effect estimation (with country and time fixed effects), we establish a positive association between participation in GAVCs and self-employment, but this is not statistically significant. Looking at the sectors, we observe some interesting insights. The upstream agricultural sector is associated with self-employment while the downstream food and beverage sector is negatively associated with self-employment. This is somewhat intuitive given the opportunities and potential inherent in many agrifood value chains activities especially in developing countries such as the horticultural chains in Namibia, Kenya, Senegal and Egypt, the tobacco value chains in Southern African countries such as Malawi and Zimbabwe, cocoa value chain in West Africa, sugar value chain

⁸ Informal output is measured by a dynamic general equilibrium model based (DGE) estimates of informal output (% of official GDP) and self-employment is measured by the percentage of self-employment in total employment. Source: Elgin, C., M. A. Kose, F. Ohnsorge, and S. Yu. 2021. “Understanding Informality”, C.E.P.R. Discussion Paper 16497, Centre for Economic Policy Research, London

in Brazil and the oil palm value chain in Southeast Asia (Maertens and Swinnen 2009; van den Broeck et al. 2017; Scoones et al. 2018; Feyaerts et al. 2020; Tabe-Ojong and Abay 2023). With regards to informal output, we find that GAVC participation is positively associated with the share of informal output in the economy (Table 9). Here, the agricultural sector is positively associated with informality while the food and beverage sector is negatively associated with informality. Just like with the case of self-employment, this is expected especially given that mostly developing countries are integrated in the agriculture sub-sector and these chains are usually not well organized as compared to the downstream processing industry that is well rooted in formality given it is the closer chain to the consumers. This also suggests that the food and beverage downstream sector is usually more regulated than the traditional agricultural sector. Given the less regulation that is usually common place with informality and self-employment, this could possibly explain the negative relationship between GAVCs and income inequality especially given that poor working conditions and other social issues that have highlighted in some of these regional and GVCs (Barrientos et al. 2011). Moreover, formality is increasingly being used as a proxy for job quality which is lower in many developing countries (Hovhannisyan et al. 2022).

Table 8: Agriculture GVC participation and informality (Informal output)

	Self-Employment (% of total)			
	(1) Self Employment	(2) Self Employment	(3) Self Employment	(4) Self Employment
GAVC (%)	0.0522 (0.0384)			
Agriculture (%)		0.175*** (0.0375)		0.249*** (0.0441)
Food (%)			-0.00711 (0.0325)	-0.121*** (0.0380)
Economy	Yes	Yes	Yes	Yes
Agriculture	Yes	Yes	Yes	Yes
Population	Yes	Yes	Yes	Yes
Demography	Yes	Yes	Yes	Yes

Trade Policy	Yes	Yes	Yes	Yes
Observations	1,739	1,739	1,739	1,739
R-squared	0.208	0.218	0.207	0.223
Number of countries	123	123	123	123
Country FE	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes

Table 9: Agriculture GVC participation, informal output and self-employment

	Informal output (% of GDP)			
	(1)	(2)	(3)	(4)
	Informal output	Informal output	Informal output	Informal output
GAVC (%)	0.0241** (0.0112)			
Agriculture (%)		0.0524*** (0.0109)		0.0869*** (0.0123)
Food (%)			-0.0263*** (0.00911)	-0.0605*** (0.0102)
Economy	Yes	Yes	Yes	Yes
Agriculture	Yes	Yes	Yes	Yes
Population	Yes	Yes	Yes	Yes
Demography	Yes	Yes	Yes	Yes
Trade Policy	Yes	Yes	Yes	Yes
Observations	2,569	2,569	2,569	2,569
R-squared	0.588	0.591	0.588	0.597
Number of countries	135	135	135	135
Country FE	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes

Note: All estimations include controls that are grouped under economy, agriculture, population, demography, and trade policy. They include Gross Domestic Production (GDP) per capita Growth, Foreign Direct Investment (in percentage of GDP), export of goods and services (in percentage of GDP), import of goods and services (in percentage of GDP), school enrollment ratio (in percentage of gross enrollment), the level of unemployment (in percentage of total labor force), inflation/consumer price index, land areas, cereals production, population ages between 0 and 14 (in percentage of total population), population aged between 15 and 64 (in percentage of total population), population of females (in percentage of total population), population of males (in percentage of total population), population density, rural population (in percentage of total population), urban population (in percentage of total population), regional trade agreement, free trade agreement, and currency union. Standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

5.4.2 Agriculture GVC participation and structural transformation

We also examine the role of structural transformation by looking at its relationship with GAVCs. To do so, we look at employment shares in the agricultural, industrial, and service sectors. We find a positive association between GAVC participation and employment in agriculture, although this is not statistically significant. We also find a negative association between GAVCs and employment shares in the industry and service sectors. Looking at the various sectors, we observe insights that are supportive of the above estimated results (Table 10). First, the agricultural sector is positively associated with employment in agriculture and negatively associated with employment shares in the industry and service sectors. This is not surprising given the nature of most agricultural GVCs which are mostly embedded in the production of agricultural products as raw materials which are then used as intermediaries to produce agricultural products which are further exported. Second, participation in the food and beverage sector is negatively associated with employment shares in agriculture and positively associated with employment shares in the industry and service sectors. These are intuitive insights suggesting the importance of the food and beverage sub-sector in stirring aspects of industrialization and structural transformation. Our insights here corroborates earlier analysis that showed that downstream participation in GAVCs is associated with structural transformation (Lim 2021).

Table 10: Agriculture GVC participation and structural change (employment share)

	(1) Employment in Agriculture (% of total)	(2) Employment in Industry (% of total)	(3) Employment in Services (% of total)
GAVC (%)	0.024 (0.019)	-0.050*** (0.013)	0.026* (0.014)
Agriculture (%)	0.133*** (0.0208)	-0.0937*** (0.0144)	-0.0391** (0.0160)
Food (%)	-0.106*** (0.0169)	0.0278** (0.0117)	0.0784*** (0.0131)
Economy	Yes	Yes	Yes

Agriculture	Yes	Yes	Yes
Population	Yes	Yes	Yes
Demography	Yes	Yes	Yes
Trade Policy	Yes	Yes	Yes

Observations	2,750	2,750	2,750
R-squared	0.587	0.397	0.752
Number of countries	144	144	144
Country FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes

Note: All estimations include controls that are grouped under economy, agriculture, population, demography, and trade policy. They include Gross Domestic Production (GDP) per capita Growth, Foreign Direct Investment (in percentage of GDP), export of goods and services (in percentage of GDP), import of goods and services (in percentage of GDP), school enrollment ratio (in percentage of gross enrollment), the level of unemployment (in percentage of total labor force), inflation/consumer price index, land areas, cereals production, population ages between 0 and 14 (in percentage of total population), population aged between 15 and 64 (in percentage of total population), population of females (in percentage of total population), population of males (in percentage of total population), population density, rural population (in percentage of total population), urban population (in percentage of total population), regional trade agreement, free trade agreement, and currency union. Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

6. Conclusion

GAVCs continue to evolve rapidly from a global perspective and have been argued to be critical in agricultural and food systems transformation. They have been shown to induce structural transformation, with significant job creation and labor market consequences. Yet, little is known about their implications for inequality, despite global concerns about inequality and whether gains from globalization always trickle down to everyone. Moreover, significant heterogeneity could exist in this relationship depending on whether countries are participating in the upstream agricultural sector or the downstream food and beverage sector. Given this, we constructed a rich and recent panel of 167 countries from 1990 to 2020 to examine the relationship between GAVCs and income inequality.

We underscore five important results. First, we establish a negative association between GAVCs and income inequality, proxied using both Gini indices and income shares. Second and related to the first, we show that this negative association is driven by the downstream food and beverage sector as opposed to the upstream agricultural sector, which is positively associated with income inequality. Third, we

consider some aspects of GAVC position, where we highlight a negative association between both the agriculture and food and beverage sector and income inequality. In the interest of exploring what may be driving these results, the fourth important insight is that GAVC participation is associated with both structural transformation and labor market employment, with some implications for job formality. These results are quite divergent and contrasting for both the agriculture GVC and the food and beverage sectors. This confirms some earlier analysis on the important role of GAVCs in spurring growth and transformation through job creation and employment effects. Finally, we uncover significant heterogeneity in the relationship between GAVCs and income inequality based on both the World Bank income-level classifications and some regional differences. These differences could be informative about the agrifood sector in other countries and regions and about the role of institutions and policies.

Our results and the insights from this analysis are relevant from both an empirical and policy level perspective. We provide insights on the income inequality implications of GAVC participation. From an empirical perspective, these are novel results that point to the role of policy in strengthening participation in GAVCs. But as the results show, there are considerable heterogeneities, pointing out that a one-size-fits all solution or a universal approach to GAVC participation may not work or may not be optimal for all countries in terms of income and welfare distributions. It is important that policy considers these heterogeneities and explores local/regional advantages to reduce such unintended income effects. For instance, while participation in both agriculture and food GVCs are associated with reductions in income inequality in Asia, participation in the food and beverage GVC is associated with a reduction in income inequality in Africa. For the latter, this could mean that significant efforts should be geared towards improving participation in African countries in the food and beverage sectors, i.e., food processing. Food processing and its associated variants in the downstream segment of GVC may imply some industrialization. This may be an important way of addressing income inequality but even without this benefit, greater investments in the food and beverage sector could significantly transform the economy with large labor market implications and employment growth. As highlighted by our

results, there could also be implications for job formality, with impacts on livelihoods and standard of living. For South America, where both the upstream and downstream agriculture and food and beverage GVC participation are associated with increases in inequality, this may reflect exclusion and poor working conditions, which may be present in both upstream and downstream components of GVCs. Addressing these labor market issues may be important in generating more equitable benefits from GAVC participation.

Despite these important results, which may be very appealing for policymakers given that About 80 percent of global trade occurs through GVCs, we mention some shortcomings and limitations of our analysis. First, our results are prudent estimates of the phenomena we analyze given some of the data limitations associated with GVC measures more broadly. Second, we refer to our estimates as associations. Although we have used rich panel data coupled with fixed effect, IV approaches, and the GMM estimator, we refrain from any strict causal interpretation of the analysis. This hesitation stems from the fact that the different identification strategies all have assumptions that in some cases can be restrictive. Our use of different estimators can lessen this, especially as we find robust estimates when analyzing country fixed effects regressions, IV regressions, and the GMM estimator. Also, we have underscored significant heterogeneity using both the income level of the countries and regional divisions. It would be important for future efforts to delve into these heterogeneities to deepen understanding of what drives them.

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